```
Loaded 6 trade logs: ['2026-02-16', '2026-02-17', '2026-02-20', '2026-02-23', '2026-02-24', '2026-02-25']
```

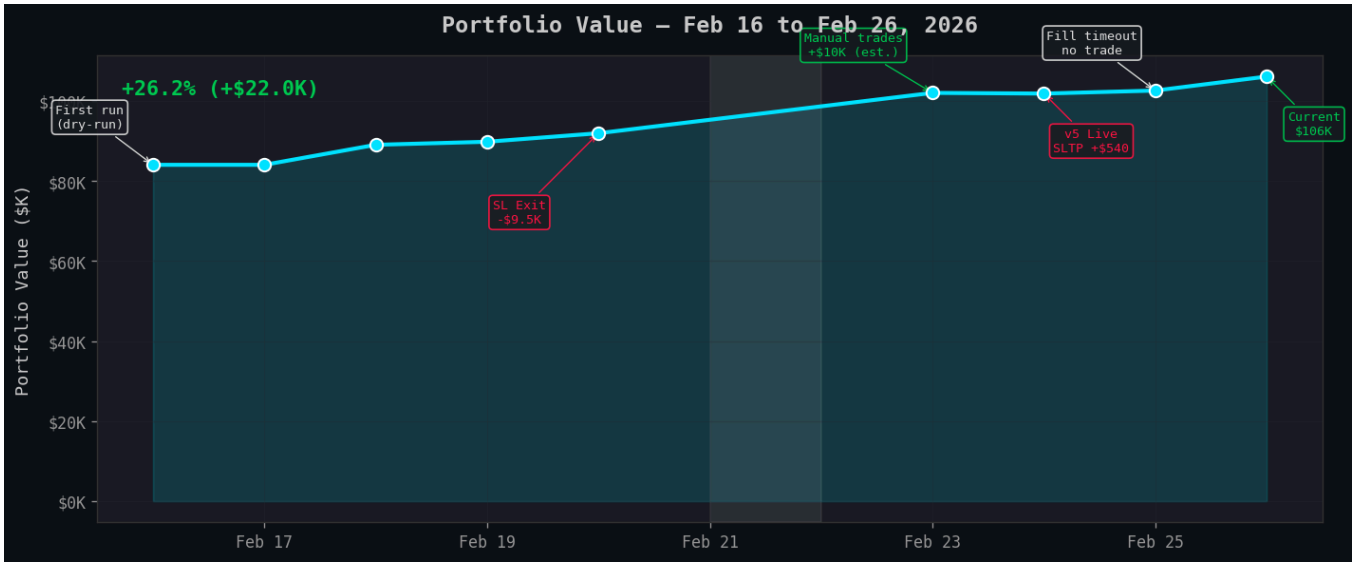
0DTE Iron Condor Bot — 4-Week Performance Report

Date Range: February 16 – February 26, 2026
Strategy: 0DTE SPX Iron Condors | 10-delta short strikes | $25 wings
ML Pipeline: v2 → v3 → v4 → v5 (evolved over 10 trading days)

Quick Stats

Metric	Value
Trading days logged	8 (6 live attempts, 1 dry-run, 1 monitor-only)
Successful fills	3 (Feb 17 manual, Feb 20, Feb 24)
Fill timeouts	1 (Feb 25 — order WORKING for 5 min, cancelled)
Stop-loss exits	1 (Feb 20 — 10% portfolio SL)
Profit-lock exits	1 (Feb 24 — SLTP 0.5% floor)
Portfolio trajectory	$84,000 \rightarrow$106,041 (+26.2%)
ML model versions used	v2 LightGBM → v3 RF → v5 RF

> **Note:** The large portfolio jump ($91K \rightarrow$102K) between Feb 20–23 is from manual trades placed directly through Schwab UI, not captured in bot logs.



Date	Mode	ML	Short Strikes	Qty	Credit	Exit	Est. P&L	Notes
Feb 16 (Mon)	Dry-Run	None (--skip-ml)	6750p / 6920c	13	—	Strike selection failed	—	First bot run; option chain missing callExpDateMap
Feb 17 (Tue)	Live	v2 LightGBM (0.82)	6780p / 6880c	30	$2.48	Order rejected	—	Paper mode bug; manual entry via Schwab UI (15+20 contracts)
Feb 18 (Wed)	Monitor	—	(manual positions)	—	—	**Manual close**	**+$5K (est.)**	Monitored manual ICs; BP collapsed $195K \rightarrow$35K during hold
Feb 19 (Thu)	Live	v3 (failed load)	—	—	—	sklearn ImportError	—	sklearn._forest module not found; scipy version conflict
Feb 20 (Fri)	Live	v3 (--skip-ml)	6845p / 6935c	42 (23 monitored)	$4.40	**STOP-LOSS**	**-$9,545**	SL at 11:22 AM; debit $8.55 (1.9x credit); 10% portfolio loss
Feb 23 (Sun)*	Dry-Run	v5 (first test)	6780p / 6895c	36	$2.50	skip_wait_test	—	*Sunday test — v5 model validation only
Feb 24 (Tue)	Live	v5 (TP25=0.675)	6830p / 6905c	36	$1.95	**SLTP 0.5%**	**+$540**	v5 entry scan → filled 11:10 → SLTP profit lock at 11:54
Feb 25 (Tue)	Live	v5 (TP25=0.653)	6910p / 6955c	36	$2.00	**FILL TIMEOUT**	—	v5 triggered at 10:05; order WORKING 5 min → cancelled; rest of day below threshold

Trade Case Studies

A. Feb 20 — Stop-Loss Event

The worst day in the period. A sharp intraday SPX move blew through the iron condor's short strike.

Detail	Value
Position	6835/6860p – 6925/6950c (23 contracts)
Entry credit	$4.40 per contract
SL threshold	$8.36 (1.9x credit = 10% portfolio)
SL warning	$6.38 (halfway to exit)
Exit debit	$8.55 at 11:22 AM ET
Estimated loss	-$9,545 (-10.4% of portfolio)
Time to SL from warning	~7 minutes (debit $6.90 \rightarrow$8.55)

Key insight: The portfolio-based SL (10%) limited damage. Under the old 4x multiplier system, the SL would have been at $17.60 — potentially a 26.7% portfolio loss.

B. Feb 24 — SLTP Profit Lock (v5 First Live Trade)

First live trade using the v5 ML model with tiered profit protection.

Detail	Value
Position	6805/6830p – 6905/6930c (36 contracts)
v5 scores at entry	TP25=0.675, TP50=0.665
Entry credit	$1.95 at 11:10 AM ET
TP target	$1.30 debit (keeping 33% of credit)
SL threshold	$4.75 debit (10% portfolio)
SLTP tier 2.5% activated	11:02 AM (debit $1.14)
SLTP tier 1.0% activated	11:46 AM (debit $1.60)
SLTP 0.5% floor breached	11:54 AM (debit $1.80 \geq floor$1.76)
Estimated profit	+$540 (+0.5% of portfolio)
Hold time	44 minutes

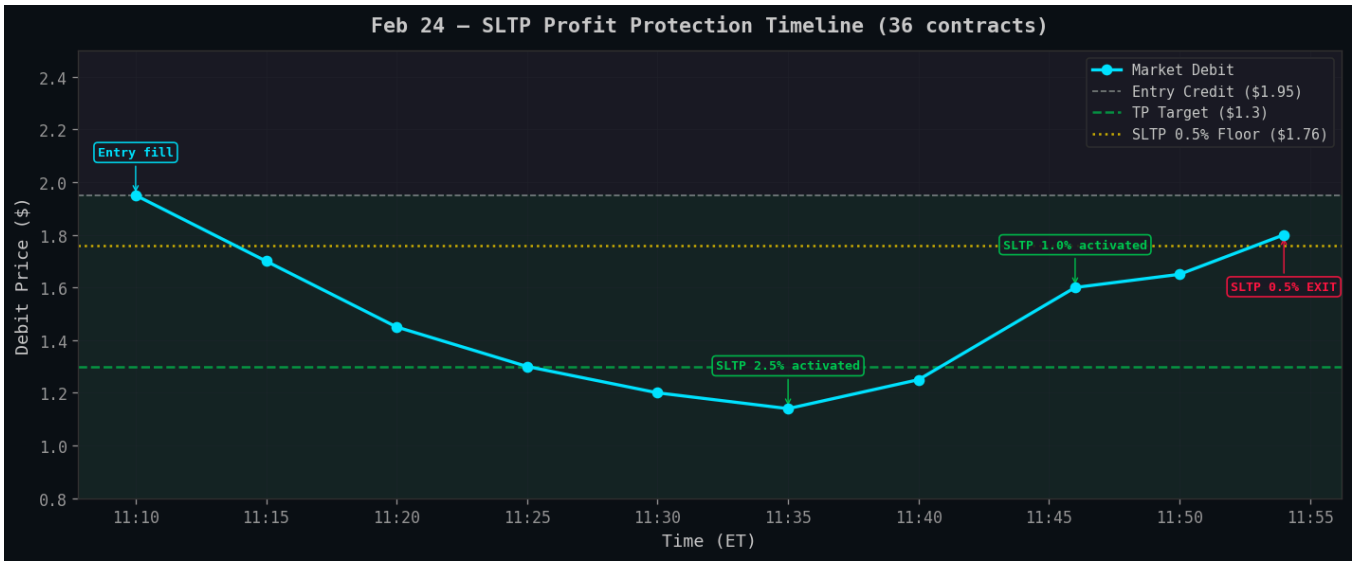
Key insight: SLTP protected gains before the full 25% TP was reached. The position briefly touched 2.5% profit territory (debit $1.14) but the market pulled back. Without SLTP, this could have turned into a flat or losing trade.

C. Feb 25 — Fill Timeout (Liquidity Miss)

v5 model triggered an entry signal but the order never filled within the 5-minute timeout window.

Detail	Value
v5 scores at trigger	TP25=0.653, TP50=0.570
Short strikes selected	6910p / 6955c
Quantity	36 contracts
Credit requested	$2.00
Order status	WORKING for 5 minutes → cancelled
Subsequent scans	All below 0.65 threshold for rest of day
Portfolio value	$102,544 (no change from trade)

Key insight: The $2.00 credit may have been too aggressive for the available liquidity. The bot's 5-minute fill timeout prevented sitting in a stale order all day. After cancellation, the v5 model never reached threshold again — the early morning window was the only opportunity, and the bot correctly identified it but couldn't fill at the requested price. Consider tightening the credit (wider limit) for better fills.



ML Model Evolution

The ML pipeline evolved through 4 major versions in 9 days, each addressing specific shortcomings:

Version	Algorithm	Target Variable	Features	AUC / CV	Training Data	Key Insight
v2	LightGBM	P(win)	35 (market+vol+regime)	—	trades.duckdb	First attempt; broad feature set, overfitting
v3	RandomForest	P(big_loss)	14 (VIX+IV+gap+range)	~0.60	763 trading days	Flipped target: skip worst 11% of days
v4	RandomForest	P(profitable @ 3PM)	24 (v3+technicals+time)	0.537	37K 5-min windows	Intraday timing; redesigned

Version	Algorithm	Target Variable	Features	AUC / CV	Training Data	Key Insight
						with 19/24 time-varying features
v5.1	RandomForest	P(hit 25% TP)	26 (v4+FOMC/CPI)	0.593 (CV 0.611)	37K 5-min windows	Matches actual dynamic TP strategy
v5.2	RandomForest	P(hit 50% TP)	26 (v4+FOMC/CPI)	0.666 (CV 0.669)	37K 5-min windows	Better separability (67.9% base rate vs 84.6%)

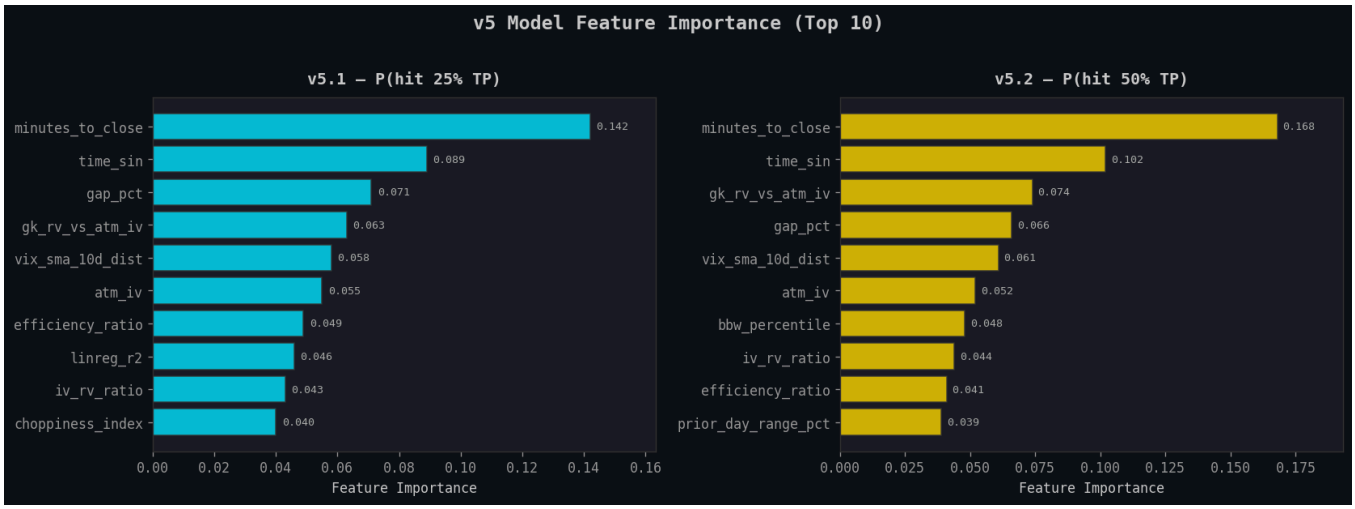
Key Design Decisions

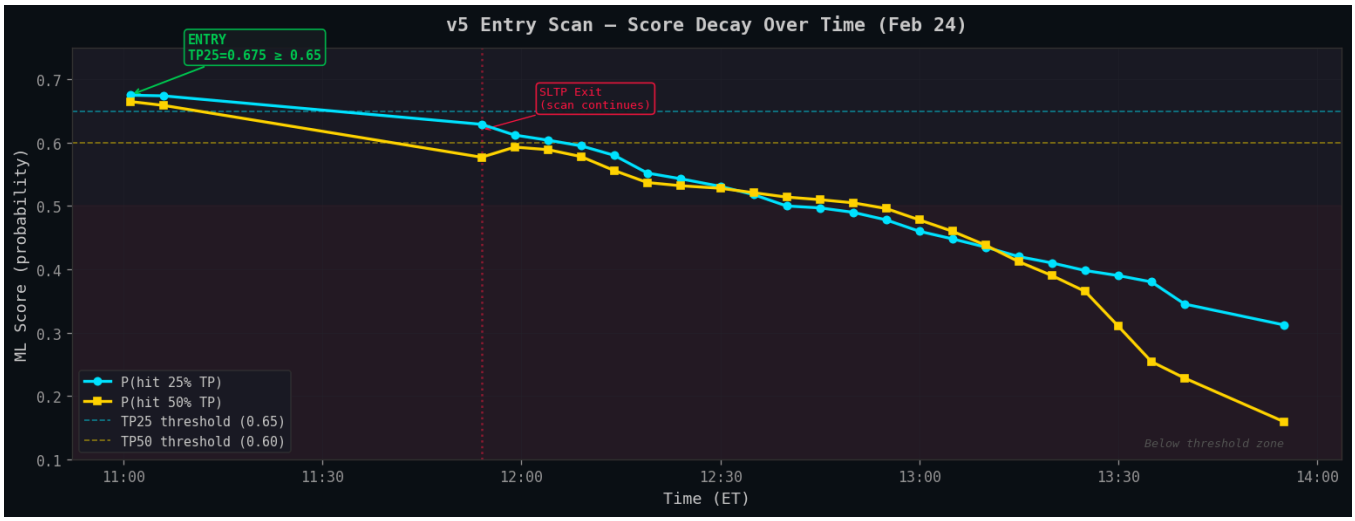
1. **v2→v3: Flip the target.** Predicting "will this win?" with 84% base rate gives no edge. Predicting "will this be a catastrophic loss?" (11% base rate) is more actionable.

2. **v3→v4: Add time-varying features.** v4's initial 14/17 features were day-level constants (same value for every 5-min window) → AUC 0.537. Redesigned with 11 technical indicators computed on rolling intraday candles → 19/24 time-varying.

3. **v4→v5: Match the target to the strategy.** v4 predicted "profitable at 3PM" but the bot uses dynamic TP (exit at 25% profit). v5 predicts P(hit 25% TP) — directly what we care about.

4. **Economic calendar.** FOMC/CPI/NFP days have distinct volatility regimes. Added `is_fomc_day` and `is_fomc_week` as binary features. Data showed FOMC days have **higher** TP hit rates (counter-intuitive), so the FOMC skip gate defaults to off.





Risk Management Framework

Three layers of protection, each operating independently:

1. Dynamic Daily-Target Take Profit

Instead of a fixed 25% TP, the bot computes the exact debit needed to hit the daily 2% target:

```
remaining_target = (portfolio × 2%) − realized_pnl_today
tp_debit = credit − (remaining_target / (qty × 100)) − fee_per_share
```

For the first trade of the day, dynamic TP ≈ fixed 25%. If a prior trade already captured some profit, the TP adjusts tighter to only capture the remaining delta.

2. Portfolio-Based Stop Loss (10%)

Max loss per trade is capped at 10% of portfolio value:

```
sl_debit = credit + (portfolio × 10% − closing_fees) / (qty × 100)
```

Scenario	Old (4x credit)	New (10% portfolio)
$2.40 credit$,89K portfolio	SL @ $9.60 (26.7% loss)	SL @ $5.07 (10% loss)
$4.40 credit$,92K portfolio	SL @ $17.60 (42.4% loss!)	SL @ $8.36 (10% loss)

3. Tiered SLTP (Trailing Profit Lock)

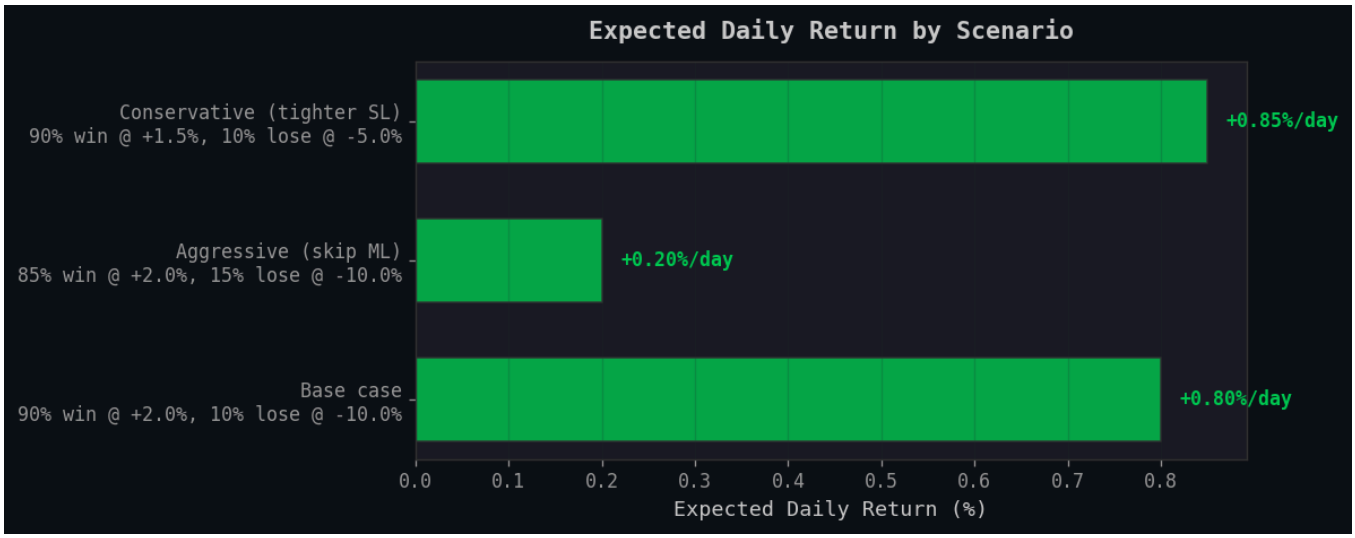
Once a position reaches a profit tier, SLTP sets a floor — if the debit rises back above the floor, the position is closed to lock in gains.

Tier	Activation (% of portfolio)	Floor (lock-in %)	Behavior
1	1.0%	0.5%	First lock — exit if profit falls to 0.5%
2	1.5%	1.0%	Ratchet up — lock at 1.0%
3	2.0%	1.5%	Daily target reached — lock at 1.5%
4	2.5%	2.0%	Bonus territory — lock at 2.0%
5	3.0%	2.5%	...
6	3.5%	3.0%	...
7	4.0%	3.5%	...
8	4.5%	4.0%	...
9	5.0%	4.5%	Max tier — nearly full profit locked

Gap between activation and floor: 0.5% — gives the position room to breathe without triggering premature exits.

Expected Value Model

Assuming the v5 model filters out the worst days:



Configuration & Architecture

Bot Settings

Setting	Value	Purpose
`BOT_SHORT_DELTA`	0.10	10-delta short strikes
`BOT_WING_WIDTH`	$25	Wing width
`BOT_DAILY_TARGET_PCT`	2%	Daily profit target

Setting	Value	Purpose
`BOT_TAKE_PROFIT_PCT`	25%	Close at 25% of credit
`BOT_STOP_LOSS_PCT`	10%	Max portfolio loss per trade
`BOT_ENTRY_TIME`	12:00 ET	Fixed-time fallback entry
`BOT_EXIT_TIME`	15:00 ET	Forced close
`BOT_V4_ENTRY_START`	10:00 ET	v4/v5 scan start
`BOT_V4_ENTRY_END`	14:00 ET	v4/v5 scan end
`BOT_V4_SCAN_INTERVAL`	5 min	Scan frequency
`BOT_V5_TP25_THRESHOLD`	0.65	v5 entry threshold
`BOT_MIN_CONFIDENCE`	0.50	v3 risk filter threshold
`BOT_COST_PER_LEG`	$0.65	Per-leg transaction cost
`BOT_MAX_CONTRACTS`	50	Hard cap
`BOT_MONITOR_INTERVAL`	5s	Position check frequency

Architecture (Unified Adaptive Mode — Feb 25 Refactor)

```
┌──────────────────────────────────────────────────────────────┐
│                zero_dte_bot.py (unified)                     │
│                                                              │
│  ┌──────────┐   ┌──────────────────────────────────┐        │
│  │Pre-flight│ → │  Detect Schwab Positions         │        │
│  │  Check   │   │  ├─ Positions? → Monitor path    │        │
│  └──────────┘   │  └─ No positions? → Entry path   │        │
│                 └──────────────────────────────────┘        │
│                              │                               │
│                              ↓                               │
│  ┌──────────────────────────────────────────────────┐       │
│  │        Entry Phase (v5→v4→fixed-time)            │       │
│  │  ML Gate (v3) → Scan Loop → Order → Fill Wait    │       │
│  └──────────────────────────────────────────────────┘       │
│                              │                               │
│                              ↓                               │
│  ┌──────────────────────────────────────────────────┐       │
│  │      Unified Monitor Loop (5s polling)           │       │
│  │  5% Ceiling TP │ Portfolio SL │ SLTP Tiers       │       │
│  │  Position change detection (rolls, manual)       │       │
│  │  3:00 PM forced close                            │       │
│  └──────────────────────────────────────────────────┘       │
└──────────────────────────────────────────────────────────────┘
                              │
                              ↓
                 ┌──────────────────────────┐
                 │    Schwab API Client     │
                 │  OAuth2 + Rate Limiter   │
                 └──────────────────────────┘
```

CLI (simplified Feb 25)

```
python scripts/zero_dte_bot.py              # Live mode (default,
prompts YES)
python scripts/zero_dte_bot.py --paper       # Paper/dry-run mode
python scripts/zero_dte_bot.py --skip-wait   # Testing: skip time
waits
python scripts/zero_dte_bot.py --skip-ml     # Testing: skip
ML/ThetaData
python scripts/zero_dte_bot.py --strikes 6910,6955  # Manual strikes
```

ML Fallback Chain

```
v5 (TP25+TP50 scores) ──[no model?]──→ v4 (profitable@3PM) ──[no
model?]──→ Fixed-time entry
     │                                        │
     └── v3 pre-screen (risk filter) ─────────┘
```

Each level falls back gracefully — removing any model file causes zero regression, just reverts to the simpler strategy.

Summary & Next Steps

Key Takeaways

1. **Portfolio-based SL is essential.** The Feb 20 stop-loss was painful (-$9.5K, -10.4%) but survivable. The old 4x-credit multiplier would have allowed a 27-42% loss on a single trade.

2. **SLTP profit protection works.** Feb 24's trade reached 2.5% profit territory but pulled back. SLTP locked in +0.5% instead of letting it turn negative.

3. **Fill quality matters.** Feb 25 showed the v5 model correctly identifying an early-morning entry window (TP25=0.653), but the order sat WORKING for 5 minutes and never filled. The $2.00 credit request may have been too aggressive for available liquidity.

4. **ML model evolution is iterative.** 4 major versions in 9 days, each fixing specific problems:

   - v2: Predicted win/loss → too noisy (84% base rate)
   - v3: Flipped to predict catastrophic loss → skip worst days
   - v4: Added intraday timing → feature engineering matters more than algorithm
   - v5: Matched target to strategy (TP hit prediction) → AUC 0.67
5. **Time is the strongest feature.** `minutes_to_close` dominates both v5 models — entering earlier in the day significantly increases probability of hitting TP.

6. **Operational failures are the biggest risk.** Three days lost to: sklearn version mismatch (Feb 19), order rejected due to paper mode bug (Feb 17), fill timeout (Feb 25). Bot reliability > model accuracy.

Portfolio Growth

- **Start:** $84,000 (Feb 16)
- **End:** $106,041 (Feb 26)
- **Change:** +$22,041 (+26.2%)
- **Bot-attributed:** ~+$540(Feb24SLTP)and-$9,545 (Feb 20 SL) = **-$9,005 net from bot**
- **Manual trades:** ~+$31,046 (placed directly through Schwab UI)

Planned Improvements

- **Fill price improvement** — widen limit price or use adaptive pricing to avoid Feb 25-style timeouts
- **Intraday VIX candles** as additional v5 features
- **GEX/DIX data** for better catastrophic-day detection
- **Multi-IC support** — enter a second IC if the first exits profitably before 2PM
- **SL slippage monitoring** — bot now polls every 5s (reduced from 30s) to catch fast reversals
- **Automated backtesting** — run v5 model against historical data to validate before live

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Report generated Feb 26, 2026. Data sources: bot JSON/log files, Schwab account snapshots. Portfolio: $106,041.